

Top 15 Azure Synapse Analytics Interview Questions and Answers

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Organizations can store and analyze massive amounts of data using Azure Synapse Analytics, a cloud-based data warehouse service. Azure Synapse Analytics is one of the most popular services for [Azure Data engineer](#) professionals. Azure Synapse Analytics integrates with other Azure data services, making it easy for data engineers to use various tools within a single platform. [Chat](#)

it is a must-have skillset in your resume if you wish to pursue a career as an Azure data engineer. This blog covers the 15 latest Azure Synapse Analytics interview questions and answers to help you ace your [data engineer interview](#).



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Top 15 Azure Synapse Analytics Interview Questions

We have divided the Azure Synapse Analytics interview questions into two sections based on your level of expertise- beginner and experienced, and one section on scenario-based questions. These interview questions will test your knowledge of the popular [Azure Synapse Analytics Service](#).

Azure Synapse Analytics Interview Questions for Beginners

1. How does Azure Synapse differ from Azure Databricks?

Azure Synapse is a [data integration](#) service with some amazing transformation capabilities while Azure Databricks is data analytics focussed platform build on top of Spark. Azure Synapse integrates [big data](#) analytics and enterprise [data warehouse](#) into a single platform. While Databricks allows customers to develop complex machine learning algorithms and perform big data analytics. However, both both Synapse Analytics and [Azure Databricks](#) can be used together when building a [data pipeline](#).

2. Define the linked service in Azure synapse Analytics.

The external sources outside the Azure Synapse Analytics workspace are connected via Linked Services. A linked service is required in a data pipeline to read or write data to source or destination. For instance, you can use it to establish a connection to the [Azure Data Lake](#) storage account to execute [SQL](#) queries on the files. Suppose you want to pull data from an on-premises server into the cloud or connect to the Azure data lake storage account to perform SQL queries on the files. In these cases, you must build a linked service by providing data such as a username, password, and server address to establish a connection.

3. What do you understand by the default SQL pool in Azure Synapse Analytics?

Anytime you build an Azure synapse analytics workspace, a serverless SQL pool is already set up. It is referred to as the built-in serverless SQL pool and is a default SQL pool that is accessible for use.

The default SQL pool in Azure Synapse Analytics helps with query optimization, data distribution, and data compression.

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4. What does the Azure Synapse Analytics OPENROWSET function do?

The OPENROWSET function lets data engineers read data from diverse data sources including flat files, RDBM's, and other OLE DB sources. In [Azure](#) Synapse, one uses the OPENROWSET function to read the file as a table. For instance, you want to perform the queries on a file saved in the ADLS account. Using the rowset function, which converts each row of a file into a row of a table, you can read this file as a table.

5. What makes Azure Synapse Analytics different from Azure Blob storage?

Large volumes of unstructured data, such as video, audio, and images, can be stored in the cloud using Azure Blob storage. You can gather, store, and analyze huge amounts of data with the help of Azure Synapse Analytics, a cloud-based data warehouse service. In a nutshell, Azure Blob Storage

majorly focusses on storing and retrieving data while Azure Synapse Analytics focusses on preparing, managing, and analyzing data.

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Scenario-based Azure Synapse Analytics Interview Questions

6. Assume you are the lead data engineer at your company, and a few spark notebooks have been run in the past few hours. You want to examine each notebook execution. Where will you find the past hour few hours' histories and retrieve the event log for them in Azure Synapse Analytics?

You must access the monitor tab in Azure Synapse Analytics. You can find the activities area in the monitor tab on the left. The Apache Spark application has a separate tab below it. The Spark history server will appear on the following page when you click on it. Simply click on that to view the whole history and to download the event log for a particular application run.

7. Suppose you are a data engineer for the XYZ company. The company wants to transfer its data from an on-site server to the Azure cloud. How will you achieve this using Azure synapse Analytics?

You must develop an integration runtime to transfer data from an on-premises server to a cloud server. This integration runtime will act as the self-hosted IR since the on-premise servers cannot be connected using the auto-resolve integration runtime, After creating the self-hosted IR, you can

use the copy function to build a pipeline that will transfer data from the on-premise server to the cloud server.

8. Suppose you are the lead data engineer at XYZ company, which is transitioning from on-premise to cloud. On one of their on-site servers, your organization holds some mission-critical data. As a result, you must ensure that the team's self-hosted IR is only accessible to specified team members. What is the best way to meet this requirement using Azure Synapse Analytics?

Open the Azure synapse analytics workspace studio. Navigate to the access control section under the monitor tab. You can grant access at the workspace item level there. Choose the IR item and the IR's name, role, and member data.

9. Assume that a different application will access the data created by your stored procedure in the SQL pool database. Daily updates of this data are necessary. What approach would you apply to address this issue?

To begin, open the Azure Synapse Analytics workspace and create the pipeline in the integration tab of the resource. The activity for the SQL pool stored procedure must be added to this pipeline. Choose the specified stored procedure from the SQL pool stored procedure activity. Schedule this pipeline by including the trigger that will run it daily.

10. Let's say you want to know how many SQL requests your team has already executed in the SQL

pool. Where can you see a list of every historical query run in the Azure Synapse Analytics workspace?

Any time you perform a query in Azure Synapse Analytics, a history log is generated. You must select the manage tab to see the SQL request. You can sort by time to see the executions and their respective results.

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Azure Synapse Analytics Interview Questions for Experienced

11. What does the Azure Synapse Analytics control node do?

The primary component of a Synapse SQL architecture is the control node. A distributed query engine is executed on the Control node to optimize and coordinate parallel queries. The Control node converts a T-SQL query into parallel queries executed against each distribution when submitted to a specific SQL pool. The DQP engine, used in serverless SQL pools, runs on the Control node and divides user queries into smaller ones that Compute nodes will process to optimize and coordinate distributed execution.

12. What is the method to create a pipeline in the Azure synapse analytics?

You can create a pipeline in Azure Synapse with the following steps:

- Open the Integrate hub in Synapse Studio.
- Click on 'Pipeline' to build a new pipeline. Open the Pipeline designer by clicking on the newly created pipeline object.
- You can start dragging the activities from the 'Activities' panel on the left into the blank space in the middle.
- You must publish your pipeline once it has been created. After that, you can test it to ensure there are no errors and verify that it works.

13. In Azure Synapse Analytics, how much data can you keep in a single column simultaneously?

The maximum size of a single column in Azure Synapse Analytics depends on the storage technology used and also on the datatype of the column. You can store up to 1 billion 2-byte Unicode characters using `nvarchar [(n | max)]`.

14. Briefly discuss the method for setting up a spark job in Azure Synapse Analytics.

The Apache Spark job must first be created under the Develop tab. You must now go to the integrate tab and build the pipeline there. Select the spark job activity and include it in the pipeline. Configure the spark job activity by choosing the exact spark job you intend to schedule. Lastly, you must schedule this pipeline, and your Spark task will be scheduled automatically. You can summarize the answer in following steps -

- i) Create a Spark Pool
- ii) Prepare the Data
- iii) Write the Spark Job
- iv) Submit the Spark Job
- v) Monitor the Spark Job using Synapse Analytics Studio
- vi) Schedule the Job

15. Mention some of the data security features offered by Synapse Analytics.

Synapse Analytics provides certain data security features for dedicated SQL pools, including Data Discovery & Classification, Dynamic Data Masking, Vulnerability Assessment, Advanced Threat Protection, and Transparent Data Encryption. Synapse Analytics supports encryption of data both in transit and at rest. Data can be encrypted using Azure Key Vault, which provides a secure way to store and manage encryption keys.

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